

Genomic Prediction

Open slides

Genomic Prediction

Genomic Prediction entails methods to predict breeding values based on phenotypic and molecular (SNP) data. The general concept is that the breeding value of an individual can be recovered by a linear regression on its genotype scores, i.e.

$$\hat{\mathbf{g}} = \mathbf{M}\hat{\mathbf{a}},$$

where $\mathbf{M}$ is a matrix of genotype scores in $\{0,1,2\}$ coding, denoting the copies of the major allele of SNP markers the individual carries, and $\hat{\mathbf{a}}$ is a vector of average allele effects for those markers.

The estimated breeding value ($\hat{\mathbf{g}}$, EBV) is then the sum of all the average allele effects for the marker alleles the individual carries.

Ridge Regression BLUP

The average allele effects can be estimated directly in a mixed effects model, treating the them as random effects:

$$\mathbf{y} = \mu + \mathbf{M}\mathbf{a} + \mathbf{e}$$

If the individuals in $\mathbf{M}$ are not representative of the general population (if they are top ranking individuals only, for example), we have to adjust the breeding values obtained from the marker effects by the population mean. In genomics, the population mean can be described by the allele frequencies in the general population:

$$p_i = \frac{\sum_{i=1}^p m_{b_i}}{n},$$

where p_i denotes the allele frequency of the general population at marker i and m_{b_i} are the marker covariates at marker i of the general population. Oftentimes, $\mathbf{M}$ is used as being representative of the general population, which is usually not the case.

In order to adjust the EBVs from a genomic prediction model to represent the deviations from the population mean, we subtract the average EBV of that population:

$$\hat{\mathbf{g}} = \mathbf{M}\hat{\mathbf{a}} - 2\mathbf{p}\hat{\mathbf{a}}$$

GBLUP - an equivalent Ridge Regression Animal Model

Instead of doing regression on the marker covariates directly, we can also generate a Gaussian Kernel from those covariates. That kernel is commonly referred to as the genomic relationship matrix, and we obtain it like this:

$$\mathbf{G} = \frac{\bar{\mathbf{M}}\bar{\mathbf{M}}'}{tr(\bar{\mathbf{M}}\bar{\mathbf{M}}')/n},$$

where $\bar{\mathbf{M}}_i = \mathbf{M}_i - 2p_i$. The centering of the genotypes guarantees that our EBVs reflect a deviation from the general population, and the scaling (division by $tr(\bar{\mathbf{M}}\bar{\mathbf{M}}')/n$) that our estimated variance components mimic an additive genetic variance.

The GBLUP model then becomes:

$$\mathbf{y} = \mu + \mathbf{g} + \mathbf{e}$$

with $\mathbf{g} \sim MVN(0, \mathbf{G}\sigma_g^2)$.

It is important to note that both models, the ridge regression marker model and GBLUP are statistically identical and result in the same EBVs. So we can choose the model that is most convenient to fit computationally.

GBLUP examples in R

First we will be preparing a common dataset for the upcoming exercises

```
library(BGLR)
library(rrBLUP)
library(ggplot2)
library(knitr)
library(lattice)

# load the "wheat" data set from BGLR
data(wheat)

# extract one phenotype
y = wheat.Y[,1]

# the marker covariates
M = wheat.X

# average allele frequencies ("p")
p = colMeans(M) / 2

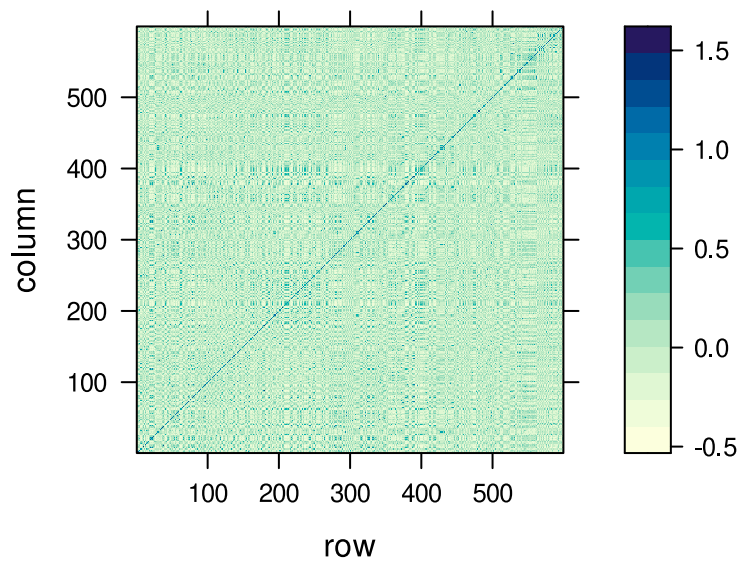
# center our Marker covariates
Mc = sweep(
  x = M,
  MARGIN = 2,
  FUN = "-",
  STATS = 2 * p
)
```

```

# compute G matrix
MMdash = Mc %*% t(Mc)
G = MMdash / mean(diag(MMdash))

# look at genomic relationships
levelplot(G)

```



With the basic dataset established, we now have many options to fit the ridge regression marker or GBLUP model.

Ridge Regression Marker Model

```

# design matrix for random effects (marker covariates become design matrix)
Z = Mc

# design matrix for fixed effects (only intercept)
X = array(1, dim = c(length(y), 1))

# fit the model with ML
mod_rr = mixed.solve(
  y = y,
  X = X,
  Z = Z,

```

```

        method = "ML"
    )

# extract marker effects
a_rr = mod_rr$u

# predict breeding values from the model
g_rr = as.numeric(Z %*% a_rr)

# extract variance components
var_a_rr = mod_rr$Vu
var_e_rr = mod_rr$Ve

# we get the "additive genetic" variance, by rescaling the variance component for
# the marker effects
var_g_rr = mod_rr$Vu * mean(diag(MMdash))

```

GBLUP Model

```

# fit the model with ML
mod_gblup = mixed.solve(
    y = y,
    K = G, # here we pass the genomic relationship matrix as kernel
    method = "ML"
)

# extract breeding values
g_gblup = mod_gblup$u

# obtain marker effects from a GBLUP model
MMdash_tmp = MMdash
diag(MMdash_tmp) = diag(MMdash_tmp) + 0.00001
a_gblup = as.numeric(t(Mc) %*% solve(MMdash_tmp, g_gblup))

# extract variance components
var_g_gblup = mod_gblup$Vu
var_e_gblup = mod_gblup$Ve

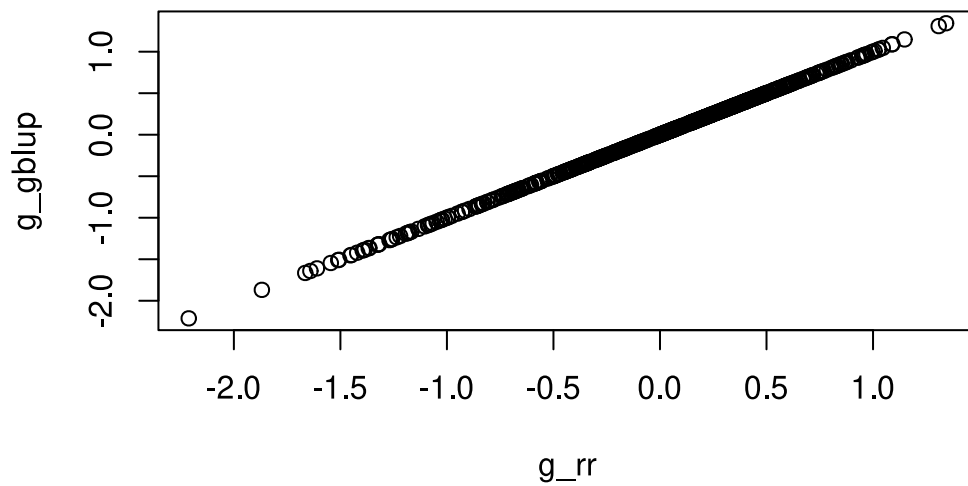
```

Check equivalence of models

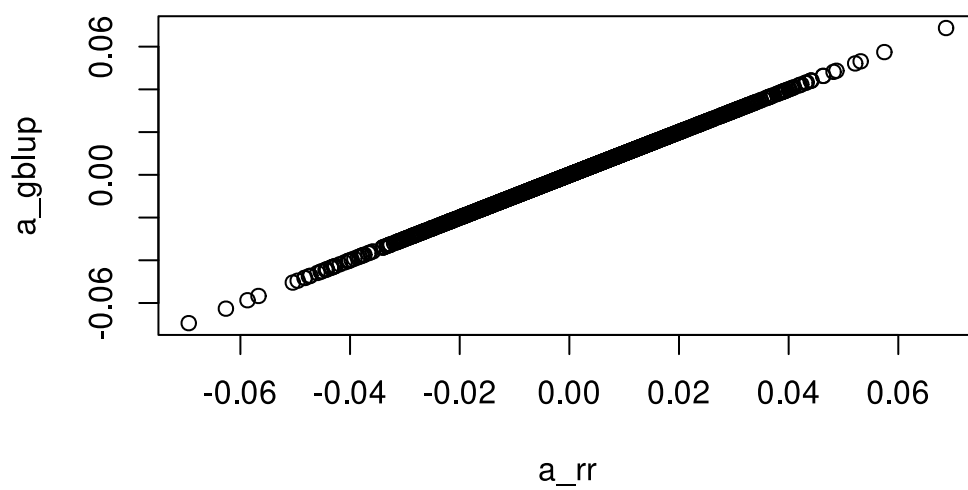
```

# plot breeding values from both models
plot(g_rr, g_gblup)

```



```
# plot marker effects from both models  
plot(a_rr, a_gblup)
```



```
# some summaries
summary(lm(g_rr ~ g_gblup))
```

```
Call:
lm(formula = g_rr ~ g_gblup)

Residuals:
      Min       1Q   Median       3Q      Max
-1.880e-06 -3.197e-07  2.300e-10  2.997e-07  1.278e-06

Coefficients:
              Estimate Std. Error  t value Pr(>|t|)
(Intercept)  4.354e-16  2.041e-08      0      1
g_gblup      1.000e+00  3.618e-08 27638903 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.994e-07 on 597 degrees of freedom
Multiple R-squared:  1, Adjusted R-squared:  1
F-statistic: 7.639e+14 on 1 and 597 DF, p-value: < 2.2e-16
```

```
summary(lm(a_rr ~ a_gblup))
```

```
Call:
lm(formula = a_rr ~ a_gblup)

Residuals:
      Min       1Q   Median       3Q      Max
-8.103e-08 -1.415e-08  2.870e-10  1.360e-08  8.062e-08

Coefficients:
              Estimate Std. Error  t value Pr(>|t|)
(Intercept)  7.978e-10  6.017e-10  1.326e+00   0.185
a_gblup      1.000e+00  3.223e-08  3.102e+07 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.15e-08 on 1277 degrees of freedom
```

```
Multiple R-squared:      1, Adjusted R-squared:      1
F-statistic: 9.624e+14 on 1 and 1277 DF,  p-value: < 2.2e-16
```

```
# variance components and heritabilities
vc = rbind(
  data.frame(
    model = "Marker Model",
    var_g = var_g_rr,
    var_e = var_e_rr,
    h2 = var_g_rr / (var_g_rr + var_e_rr)
  ),
  data.frame(
    model = "GBLUP",
    var_g = var_g_gblup,
    var_e = var_e_gblup,
    h2 = var_g_gblup / (var_g_gblup + var_e_gblup)
  )
)

kable(vc, digits = 3)
```

model	var_g	var_e	h2
Marker Model	0.605	0.539	0.529
GBLUP	0.605	0.539	0.529

Cross Validation

In order to assess the performance of our genomic prediction models, we can run a cross validation scheme, in which we train marker effects with a subset of the data and predict the EBVs of the left out samples based on the training model.

```
# define function to create cross validation samples
cCV <- function(y, folds=5, reps=1, matrix=FALSE, seed=NULL) {

  if(!is.vector(y)) stop("y must be a vector")
  if(folds < 2) stop("'folds' must be larger than one")
  if(reps < 1) reps = 1

  reps = as.integer(reps)
  folds = as.integer(folds)

  isy <- (1:length(y))[!is.na(y)]
```

```

n_vec <- rep(NA,folds)
start <- rep(NA,folds)
end <- rep(NA,folds)

n_vec[1:(folds-1)] <- round(length(isy)/folds,digits=0)
n_vec[folds] <- length(isy) - sum(n_vec[1:(folds-1)])

for(j in 1:folds) {

  if(j==1) {

    start[j] = 1

  } else {

    start[j] = end[j-1]+1

  }

  end[j] = sum(n_vec[1:j])
}

count=0
cv_pheno <- list(folds*reps)

if(missing(seed)) seed = as.integer((as.double(Sys.time())*1000+Sys.getpid())
%% 2^31)
set.seed(seed)

for(i in 1:reps) {

  ids <- sample(isy,length(isy),replace=F)

  for(j in 1:folds) {

    count=count+1
    y_temp = y
    y_temp[ids[start[j]:end[j]]] <- NA
    cv_pheno[[count]] <- y_temp

  }
}

```



```

}

if(matrix) {

  return(matrix(unlist(cv_pheno),nrow=length(y),ncol=length(cv_pheno),byrow=FALSE))

} else {

  return(cv_pheno)

}

}

# create 5 fold cross validation samples
cvs = cCV(y, folds = 5, reps = 10)

# run cross validation
pred = rep(as.numeric(NA), length(y))
for(i in 1:length(cvs)) pred[is.na(cvs[[i]])] = mixed.solve(y = cvs[[i]], K
= G)$u[is.na(cvs[[i]])]

# if we have more than 1 repition, need to use this
preds = list()
for(i in 1:length(cvs)) {

  this_pred = mixed.solve(y = cvs[[i]], K = G)$u[is.na(cvs[[i]])]
  this_y = y[is.na(cvs[[i]])]
  preds[[i]] = data.frame(pred = this_pred, y = this_y)

}

cors = unlist(lapply(preds, function(x) cor(x$pred, x$y)))

# Combine all predictions into a single vector if needed
pred <- unlist(lapply(preds, function(x) x$pred))

# plot predices vs observed
plot(pred, y)

```

```
# prediction accuracy
cor(pred, y)

# check bias
summary(lm(pred ~ y))
```

Bayesian Alphabet

A linear mixed model with a series of independent random effects has the following form:

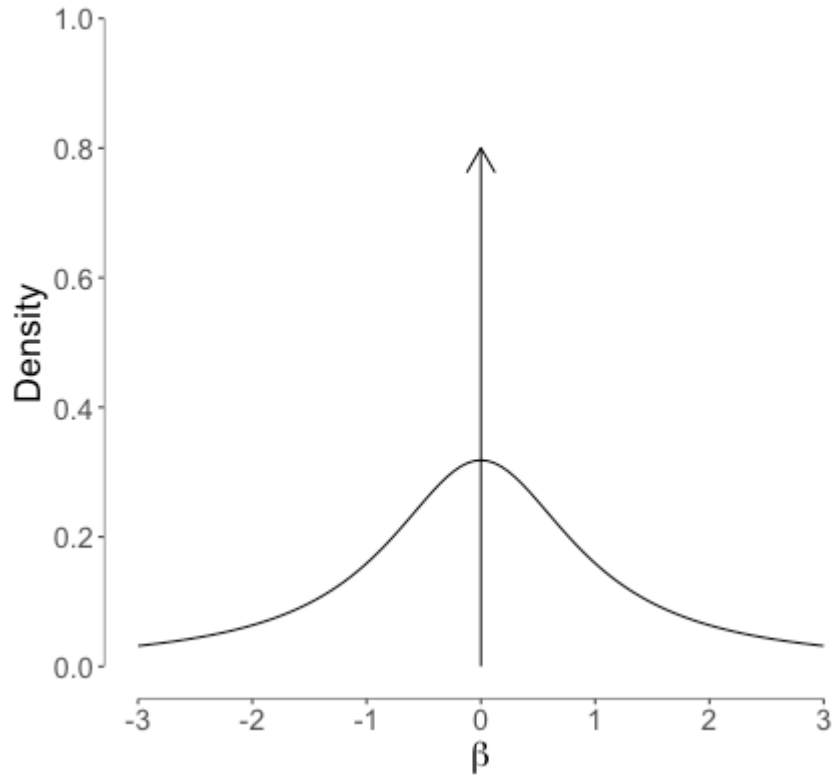
$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}_1\mathbf{u}_1 + \mathbf{Z}_2\mathbf{u}_2 + \mathbf{Z}_3\mathbf{u}_3 + \dots + \mathbf{Z}_k\mathbf{u}_k + \mathbf{e},$$

with the joint posterior density

$$\begin{aligned} p(\boldsymbol{\beta}, \sigma_{\boldsymbol{\beta}}^2, \mathbf{u}_1, \sigma_{u_1}^2, \dots, \mathbf{u}_k, \sigma_{u_k}^2, \sigma_{\epsilon}^2 \mid \mathbf{y}) &\propto \left\{ \prod_{i=1}^n N(y_i \mid \mathbf{x}'_i \boldsymbol{\beta} + \sum_{j=1}^k \mathbf{z}'_{ji} \mathbf{u}_j, \sigma_{\epsilon}^2) \right\} \\ &\times N(\boldsymbol{\beta} \mid 0, \mathbf{I} \sigma_{\boldsymbol{\beta}}^2) \prod_{j=1}^k N(\mathbf{u}_j \mid 0, \mathbf{I} \sigma_j^2) \\ &\times \prod_{j=1}^k \chi^{-2}(\sigma_j^2 \mid S_j, df_j) \chi^{-2}(\sigma_{\epsilon}^2 \mid S_{\epsilon}, df_{\epsilon}). \end{aligned}$$

In the Bayesian Alphabet, the prior distributions on the marker effects is altered to generally generate a sparsity pattern on the marker effects, meaning feature selection. This feature selection is achieved by using *spike-and-slap* mixture priors. Those priors have a point mass at zero and a usually normal slap. Noisance marker effects ideally get absorbed in the point mass at zero, whereas relevant markers have a higher mixture probability for the slap, and get a significant effect size.

Spike-and-Slab Prior



{<https://fabindablander.com/r/Spike-and-Slab.html>}

Table 1. Summary of some effect size distributions that have been proposed for polygenic modeling.

Effect Size Distribution		Keywords	Selected References
Name	Formula		
t	$\beta_i \sim t(0, v, \sigma_a^2)$	BayesA	[27,32,33,40]
point-t	$\beta_i \sim \pi t(0, v, \sigma_a^2) + (1 - \pi)\delta_0$	BayesB, BayesD, BayesD π	[27,32–34,40]
t mixture	$\beta_i \sim \pi t(0, v, \sigma_a^2) + (1 - \pi)t(0, v, 0.01\sigma_a^2)$	BayesC	[32,33]
point-normal	$\beta_i \sim \pi N(0, \sigma_a^2) + (1 - \pi)\delta_0$	BayesC, BayesC π , BVSR	[18,19,34]
double exponential	$\beta_i \sim \text{DE}(0, \theta)$	Bayesian Lasso	[28,39,68]
point-normal mixture	$\beta_i \sim \pi_1 N(0, \sigma_a^2) + \pi_2 N(0, 0.1\sigma_a^2) + \pi_3 N(0, 0.01\sigma_a^2) + (1 - \pi_1 - \pi_2 - \pi_3)\delta_0$	BayesR	[35]
normal	$\beta_i \sim N(0, \sigma_a^2)$	LMM, BLUP, Ridge Regression	[22,26,28,48]
normal-exponential-gamma	$\beta_i \sim \text{NEG}(0, \kappa, \theta)$	NEG	[16]
normal mixture	$\beta_i \sim \pi N(0, \sigma_a^2 + \sigma_b^2) + (1 - \pi)N(0, \sigma_b^2)$	BSLMM	Present Work

The reference list contains only a selection of relevant publications. Abbreviations: DE denotes double exponential distribution, NEG denotes normal exponential gamma distribution, and other abbreviations can be found in the main text. In the scaled t-distribution, v and σ_a^2 are the degree of freedom parameter and scale parameter, respectively. In the DE distribution, θ is the scale parameter. In the NEG distribution, κ and θ are the shape and scale parameters, respectively. Notes: 1. Some applications of these methods combine a particular effect size distribution with a random effects term, with covariance matrix K , to capture sample structure or relatedness. If $K \propto XX^T$ then this is equivalent to adding a normal distribution to the effect size distribution. The listed effect size distributions in this table do not include this additional normal component. 2. BayesC has been used to refer to models with different effect size distributions in different papers. 3. In some papers, keywords listed here have been used to refer to fitting techniques rather than effect size distributions.

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{https://xzlab.org/papers/2013_Zhouetal_PG.pdf}

BayesC π - Threshold Model

Binary Traits (0/1) can be modelled with generalized linear models in which the expectation of the observations are mapped to a linear predictor comprised of random and fixed effects. The likelihood for the data is a Bernoulli distribution in our case, with success probably linked to the linear predictor through the probit function:

$$\begin{aligned} \mathbf{y} &\sim \mathbf{B}(1, \mathbf{p}) \\ \mathbf{p} &= \text{probit}(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}) \\ \boldsymbol{\beta} &\sim \mathbf{U}(-Inf, Inf) \\ \mathbf{u} &\sim \pi N(0, \sigma_u^2) + (1 - \pi)\delta_0 \\ \pi &\sim U(0, 1) \\ \sigma_u^2 &\sim \text{Gamma}(1, 0.01) \end{aligned}$$

The following shows an implementation of that model in Jags:

```
if (!require("rjags")) install.packages("rjags")
library(rjags)

BayesCJags <- function(y, Z, threshold = FALSE, niter = 5000, burnin = 2500,
  betaPriorAlpha = 2, betaPriorBeta = 5) {

  #####
  ### First the model strings ###
  #####

  # probit threshold model
  modelstringThreshold = "
  model {
    for (i in 1:n) {
      y[i] ~ dbern(prob[i])
      probit(prob[i]) <- alpha + inprod(X[i,],beta)
    }
    for (j in 1:p) {
      ind[j] ~ dbern(pind)
      betaT[j] ~ dnorm(0,tau)
      beta[j] <- ind[j] * betaT[j]
    }
    tau ~ dgamma(1,0.001)
    alpha ~ dnorm(0,0.0001)
```

```

    # pind ~ dunif(0,1)
    pind ~ dbeta(betaPriorAlpha, betaPriorBeta)
  }
  "

# and the gaussian model
modelstringGaussian = "
model {
  for (i in 1:n) {
    y[i] ~ dnorm(alpha + inprod(X[i,],beta), tauE)
  }
  for (j in 1:p) {
    ind[j] ~ dbern(pind)
    betaT[j] ~ dnorm(0,tauB)
    beta[j] <- ind[j] * betaT[j]
  }
  tauB ~ dgamma(1,0.001)
  alpha ~ dnorm(0,0.0001)
  # pind ~ dunif(0,1)
  pind ~ dbeta(betaPriorAlpha, betaPriorBeta)
  tauE ~ dgamma(1, 0.001)
}
"

# the default (gaussian)
modelstring <- modelstringGaussian

if(threshold) {

  modelstring = modelstringThreshold

}

# some checks
if(!(is.numeric(y) | is.integer(y) | is.vector(y))) stop("y must be a numeric/
integer vector")
if(threshold & length(unique(y)) > 2) stop("y must be 0/1 for the threshold
model")

if(!is.matrix(Z)) stop("Z must be a numeric matrix")
if(nrow(Z) != length(y)) stop("Z must have as many rows as elements in y")

```

```

# make the data.frame
n = length(y)
p <- ncol(Z)

data = list(y = y, X=Z ,n = n,p = p, betaPriorAlpha = betaPriorAlpha,
betaPriorBeta = betaPriorBeta)
init = list(alpha = 0, betaT = rep(0,p), pind = 0, ind = rep(0,p), tauB = 1)
if(!threshold) init$tauE = 1

load.module("glm")
model = jags.model(textConnection(modelstring),
  data = data, inits = init)

# burnin
update(model, n.iter = burnin)

# sampling
output = coda.samples(model = model,
  variable.names = names(init),
  n.iter = (niter - burnin), thin = 1)

return(output)
}

# create "binary" trait from our phenotypes
mu_y = mean(y)
y_binary = rep(0, length(y))
y_binary[y > mu_y] = 1

Mc_small = Mc[, 1:50]

# run model
mod_bcpi <- BayesCJags(y = y_binary, Z = Mc_small, threshold = TRUE, niter =
500, burnin = 250, betaPriorAlpha = 2, betaPriorBeta = 5)

# gaussian response
mod_bcpi_gaussian <- BayesCJags(y = y, Z = Mc_small, threshold = FALSE, niter =
500, burnin = 250, betaPriorAlpha = 2, betaPriorBeta = 5)

```